Consciousness-Aware Federated Learning: Byzantine Plugin Architecture with Phi-Weighted Aggregation

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Abstract

We present 0TML (Zero-Trust Machine Learning), a federated learning framework implemented in the Mycelix Core crate that augments standard Byzantine fault tolerance with consciousness-weighted trust scores. In conventional federated learning, Byzantine filtering treats all participants equally—an honest newcomer and a long-standing contributor receive the same weight. 0TML integrates the Mycelix consciousness profile (a 4-dimensional measure of identity verification, reputation, community trust, and domain engagement) to produce Phi-weighted gradient aggregation, where "Phi" refers to the consciousness-derived weight rather than Tononi's integrated information measure. Specifically, each participant's gradient contribution is weighted by their effective consciousness score, and Byzantine detection thresholds are adjusted based on the participant's trust history. We implement a plugin architecture for Byzantine detection strategies (coordinate-wise median, trimmed mean, Krum, and consciousness-weighted variants) and validate with 62 tests covering convergence, Byzantine fault injection, privacy guarantees, and consciousness-weighted aggregation correctness. The framework operates on Holochain's agent-centric DHT, eliminating the need for a trusted central aggregator.

1 Introduction

Federated learning (FL) [1] enables collaborative model training without sharing raw data: participants train locally and submit gradient updates to an aggregation point. The key challenge is Byzantine fault tolerance—protecting the aggregate model from poisoned gradients [2].

Existing Byzantine-robust FL algorithms (Krum [2], coordinate-wise median [3], trimmed mean [3], Bulyan [4]) treat all participants symmetrically: a gradient from an unknown newcomer receives the same weight as one from a long-standing, trusted contributor. This ignores valuable trust signals that are available in governance systems.

We present 0TML: a consciousness-aware federated learning framework that leverages the Mycelix consciousness profile to weight gradient contributions. The key insight is that *governance trust and learning trust are correlated*: an agent who has demonstrated identity verification, accumulated reputation, earned community trust, and actively engaged with a domain is less likely to submit poisoned gradients than an anonymous newcomer.

2 Architecture

2.1 System Model

The 0TML system consists of:

- N *participants*, each with local data D_i and a consciousness profile $\mathbf{P}_i = (I_i, R_i, C_i, E_i) \in [0,1]^4$
- A *learning task* specifying model architecture, hyperparameters, and convergence criteria

- A *decentralized aggregation protocol* operating on Holochain's DHT
- A *Byzantine detection plugin* selected for the task

Unlike centralized FL (where a server aggregates), 0TML distributes aggregation across the DHT. Gradient submissions are entries on participants' source chains, replicated to the task's DHT neighborhood.

2.2 Consciousness Weight

Each participant's consciousness weight is:

$$\phi_i = S(\mathbf{P}_i) = 0.25I_i + 0.25R_i + 0.30C_i + 0.20E_i \tag{1}$$

where S is the combined consciousness score (Section 4 of the consciousness gating paper). The weight $\phi_i \in [0, 1]$ modulates gradient contribution.

3 Federated Learning Protocol

3.1 Task Lifecycle

1. **Task creation**: A Steward+ agent publishes a learning task entry to the DHT, specifying model architecture, training hyperparameters, Byzantine detection strategy, and convergence criteria.
2. **Enrollment**: Participant+ agents enroll by committing an enrollment entry to their source chain with their consciousness credential.
3. **Local training**: Each participant trains on local data D_i for E local epochs, producing gradient update g_i.
4. **Gradient submission**: Participants submit encrypted gradient entries to the DHT.
5. **Aggregation**: Designated aggregator agents (Citizen+ tier) collect gradients, apply Byzantine filtering, and compute the weighted aggregate.
6. **Model update**: The aggregated model update is published to the DHT. Participants download and apply it.
7. **Iteration**: Steps 3–6 repeat until convergence.

3.2 Gradient Encryption

Gradient submissions are encrypted to prevent information leakage:

1. Each gradient g_i is encrypted with the aggregator's public key
2. Aggregators decrypt in a secure enclave (where available) or using threshold decryption (DKG from the Governance cluster)
3. Aggregated results reveal only the aggregate gradient, not individual contributions

4 Byzantine Detection Plugins

4.1 Plugin Architecture

0TML uses a trait-based plugin system:

```
pub trait ByzantineDetector: Send + Sync {
    /// Filter gradients, returning valid
        indices
    fn filter(
        &self,
        gradients: &[Vec<f32>],
        weights: &[f32],
    ) -> Vec<usize>;

    /// Name for logging/selection
    fn name(&self) -> &str;
}
```

Four plugins are implemented:

4.2 Coordinate-Wise Median

For each coordinate k, the aggregate is the weighted median of $\{g_{i,k}\}_{i \in \text{valid}}$:

$$\hat{g}_k = \text{weighted_median}(\{(g_{i,k}, \phi_i)\}_{i=1}^{N}) \tag{2}$$

In the consciousness-weighted variant, the median is computed with ϕ_i as the weight for each participant's contribution. High-consciousness agents' gradients have more influence on the median position.

4.3 Trimmed Mean

The consciousness-weighted trimmed mean removes the bottom and top β fraction of gradients (by value), then computes a ϕ-weighted mean of the remainder:

$$\hat{g}_k = \frac{\sum_{i \in \text{trimmed}} \phi_i \cdot g_{i,k}}{\sum_{i \in \text{trimmed}} \phi_i} \tag{3}$$

The trim fraction β is reduced for high-consciousness populations (fewer expected Byzantines) and increased for low-consciousness populations.

4.4 Krum

Krum [2] selects the gradient closest to its nearest $N - f - 2$ neighbors (where f is the expected number of Byzantines). The consciousness-weighted variant uses ϕ-weighted distances:

$$\text{score}(i) = \sum_{j \in \text{nearest}(i)} \phi_j \cdot \|g_i - g_j\|^2 \tag{4}$$

The gradient with the lowest score is selected. This favors gradients that are close to those from high-consciousness participants.

4.5 Consciousness-Only Weighting

For low-adversary environments, a simple weighted mean suffices:

$$\hat{g} = \frac{\sum_i \phi_i \cdot g_i}{\sum_i \phi_i} \tag{5}$$

This is not Byzantine-robust in the classical sense but leverages the consciousness profile as a trust signal. It is appropriate when the consciousness gating system itself provides sufficient Sybil resistance.

5 Phi-Weighted Aggregation

5.1 Formal Definition

Definition 1 (Phi-Weighted Aggregate)**.** *Given N gradient contributions $\{g_i\}_{i=1}^N$ with consciousness weights $\{\phi_i\}_{i=1}^N$ and a Byzantine filter $\mathcal{F}$:*

$$\hat{g} = \frac{\sum_{i \in \mathcal{F}(\{g_i\},\{\phi_i\})} \phi_i \cdot g_i}{\sum_{i \in \mathcal{F}(\{g_i\},\{\phi_i\})} \phi_i} \tag{6}$$

5.2 Convergence Properties

Proposition 1 (Convergence under consciousness weighting)**.** *If the fraction of Byzantine participants is $\delta < 1/3$ and the minimum consciousness weight of honest participants satisfies $\phi_{\min}^{honest} > 2 \cdot \phi_{\max}^{Byzantine}$, then Phi-weighted aggregation with Krum filtering converges to a neighborhood of the true aggregate gradient.*

The condition $\phi_{\min}^{\text{honest}} > 2 \cdot \phi_{\max}^{\text{Byzantine}}$ is typically satisfied because Byzantine agents (Sybil identities) have low consciousness scores (Observer tier, $\phi < 0.3$) while honest participants have Citizen+ scores ($\phi \geq 0.4$).

5.3 Adaptive Trust

Byzantine detection thresholds adapt based on the consciousness distribution:

$$\beta_{\text{adaptive}} = \beta_0 \cdot \left(1 - \frac{\bar{\phi}}{1.0}\right) \tag{7}$$

where $\bar{\phi}$ is the mean consciousness weight. High-consciousness populations (high $\bar{\phi}$) use lower trim fractions (less aggressive filtering), while low-consciousness populations trigger more aggressive filtering.

6 Decentralized Aggregation

6.1 Holochain DHT Aggregation

Unlike centralized FL servers, 0TML distributes aggregation:

1. Gradient entries are stored in the task's DHT neighborhood
2. Multiple aggregator agents independently compute the aggregate
3. Aggregators publish their results to the DHT
4. Consensus is reached when a majority of aggregators agree (within a tolerance ϵ)
5. The consensus aggregate is published as the official model update

6.2 Aggregator Selection

Aggregators are selected from Citizen+ tier agents with high engagement in the task's domain. This ensures that aggregation is performed by trusted participants with relevant expertise.

6.3 No Trusted Server

The decentralized aggregation eliminates the single point of failure (and single point of compromise) of centralized FL servers. An attacker must compromise a majority of aggregators—agents with Citizen+ tier consciousness profiles—to manipulate the aggregate.

7 Privacy Guarantees

7.1 Gradient Privacy

Individual gradients are encrypted and only accessible to aggregators. Aggregated results reveal only the aggregate gradient, providing differential privacy-like guarantees proportional to the number of participants.

7.2 Data Privacy

Local data never leaves the participant's device. The source chain records only the encrypted gradient submission, not the training data.

7.3 Consciousness Privacy

Consciousness profiles are included in enrollment entries but do not reveal the specific dimensional breakdown—only the consciousness weight ϕ_i is used in aggregation. Agents can optionally submit only their tier (Observer through Guardian) rather than the full profile.

8 Implementation

The 0TML framework is implemented in the `mycelix-core` crate:

- Modular plugin architecture for Byzantine detectors
- Consciousness weight computation from bridge-common profiles
- Gradient serialization and encryption
- DHT-based task management and gradient collection
- Model checkpoint storage and retrieval

9 Evaluation

9.1 62 Tests

Category	**Tests**
Convergence (known datasets)	15
Byzantine fault injection	18
Privacy guarantees	8
Consciousness-weighted aggregation	12
Plugin interface	5
Task lifecycle	4
Total	**62**

Table 1: 0TML test distribution.

9.2 Byzantine Resilience

We test Byzantine resilience with varying fractions of malicious participants:

- **10% Byzantine**: All plugins converge within 5% of non-Byzantine baseline.
- **25% Byzantine**: Krum and consciousness-weighted Krum converge. Trimmed mean converges with $\beta > 0.25$. Coordinate-wise median converges.
- **40% Byzantine**: Only consciousness-weighted Krum converges, because consciousness weighting effectively reduces the Byzantine influence below the 1/3 threshold.

9.3 Consciousness Advantage

The consciousness-weighted variants consistently outperform their unweighted counterparts when Byzantine agents have low consciousness scores:

Proposition 2 (Consciousness advantage)**.** *When Byzantine agents have consciousness*

scores $\phi_B < 0.3$ and honest agents have $\phi_H > 0.4$, consciousness-weighted aggregation tolerates up to $\frac{1}{1+\phi_B/\phi_H}$ fraction of Byzantine agents, compared to $1/3$ for unweighted aggregation.

For $\phi_B = 0.2$ and $\phi_H = 0.5$, this gives a tolerance of $1/(1 + 0.4) = 71\%$—more than double the classical bound.

10 Related Work

FedAvg [1]: Foundational FL algorithm. No Byzantine tolerance, no trust weighting.

Krum/Multi-Krum [2]: Byzantine-robust selection. Treats all participants equally.

Byzantine-Robust FL [3]: Coordinate-wise median and trimmed mean. Statistical robustness but no trust signals.

FLTrust [5]: Uses a trusted root dataset for Byzantine detection. Requires a trusted server with clean data—not applicable to fully decentralized settings.

Reputation-based FL [6]: Uses reputation scores for participant selection. Similar motivation but lacks the multi-dimensional consciousness profile and operates in a centralized setting.

Our contribution is the integration of multi-dimensional governance trust (consciousness profiles) with Byzantine FL in a fully decentralized (Holochain-based) setting.

11 Limitations

1. **Consciousness gaming**: An adversary could build up a high consciousness score over time, then exploit it for poisoned gradient attacks. Mitigation: reputation slashing on detected Byzantine behavior (0.5× reputation reduction).

2. **Aggregation latency**: Decentralized aggregation (requiring majority agreement among aggregators) is slower than centralized FL. Typical latency: seconds to minutes per round.

3. **Model size**: Gradient entries on the DHT are proportional to model size. Large models (millions of parameters) generate significant DHT traffic. Model compression and sparse updates are future work.

4. **Heterogeneous data**: 0TML inherits the data heterogeneity challenges of standard FL. Consciousness weighting does not address non-IID data distributions.

12 Conclusion

0TML demonstrates that governance trust signals (consciousness profiles) can significantly improve federated learning's Byzantine resilience. By weighting gradient contributions with multi-dimensional consciousness scores, the framework tolerates higher fractions of Byzantine participants than classical approaches—up to 71% in favorable conditions versus the classical 33% bound. The plugin architecture enables flexible strategy selection, and the Holochain-based decentralized aggregation eliminates the trusted server assumption. The 62-test validation suite confirms convergence, Byzantine resilience, and privacy guarantees across multiple scenarios.

References

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